КН 115

Лозовий Олег

Варіант: 8

1	B
2	A_2
3	Г
4	B
5	4
6	0,225
7	$\frac{1}{6}$
8	$\sqrt{e}$
9	1,5
10	4
11	D
12	D
13	$-\infty$
14	e^{-2}
15	Вертикальні: $x=-3$ та $x=3$ Горизонтальні: немає; Похилі: $y=5x$
16	4,5

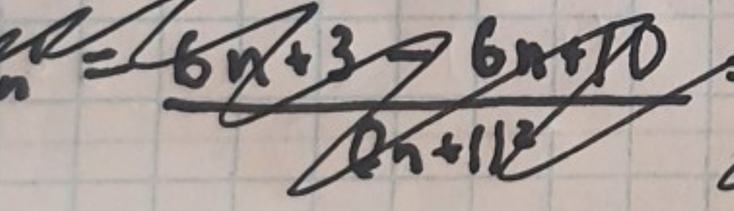
1) $e^{i\pi} = \cos\pi + i\sin\pi$

$Re(e^{i\pi}) = \cos\pi = -1 \Rightarrow$ B)

2)

$x_n = \frac{n^2}{n^2+2}$

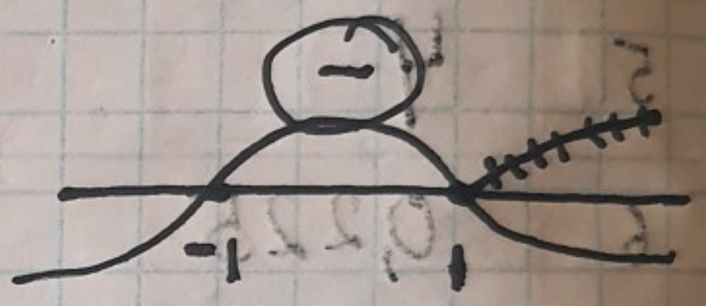
$x_n' = \frac{2n(n^2+2) - 2n\cdot n^2}{(n^2+2)^2} = \frac{2n^3 + 4n - 2n^3}{(n^2+2)^2} = \frac{4n}{(n^2+2)^2}$



$\Rightarrow$ (A)

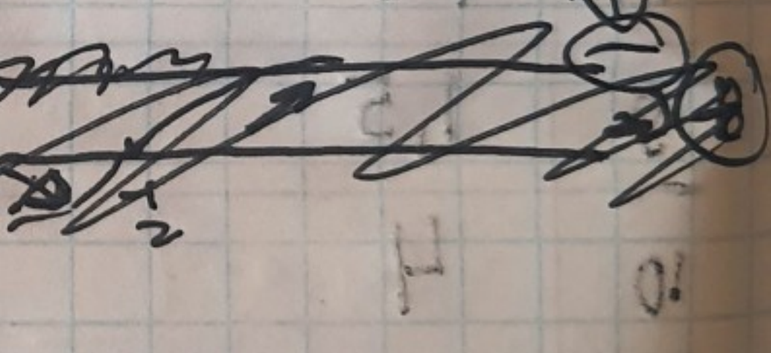
$x_n = \frac{2n}{n^2+1}$

$x_n' = \frac{2n^2+2-4n^2}{(n^2+1)^2} = \frac{-2n^2+2}{(n^2+1)^2}$



$x_n = \frac{3n+5}{2n+1}$ $n=1 \Rightarrow x_n = \frac{8}{3} = 2(6)$ $n=2 \Rightarrow x_n = \frac{11}{5} = 2,2$

$x_n' = \frac{6n+3-6n-10}{(2n+1)^2} = \frac{-13}{(2n+1)^2}$



3) $x_n = (-1)^n + 1$ $n=1 \Rightarrow x_n = 0$

$n=2 \Rightarrow x_n = 2$

$n=3 \Rightarrow x_n = 0$

$\Rightarrow$ (-)

$x_n = 2^{-n}$ $n=1 \Rightarrow x_n = 0,5$

$n=2 \Rightarrow x_n = 0,25$

$\Rightarrow$ (-)

3) $|z| = \sqrt{x^2+y^2}$

$|-z| = \sqrt{(-x)^2+(-y)^2} = \sqrt{x^2+y^2}$ $\Rightarrow$ (+)

$z = \mathrm{Re}\,z + \mathrm{Im}\,z$

$\bar{z} = \mathrm{Re}\,\bar{z} + \mathrm{Im}\,\bar{z}$

$z\bar{z} = x^2+y^2 = (\mathrm{Re}\,z)^2 + (\mathrm{Re}\,\bar{z})^2 \Rightarrow$ (−)

$z = a+bi$

$a = |a| \Rightarrow a = a \Rightarrow$ (+) (1)

$-z = -a-bi$

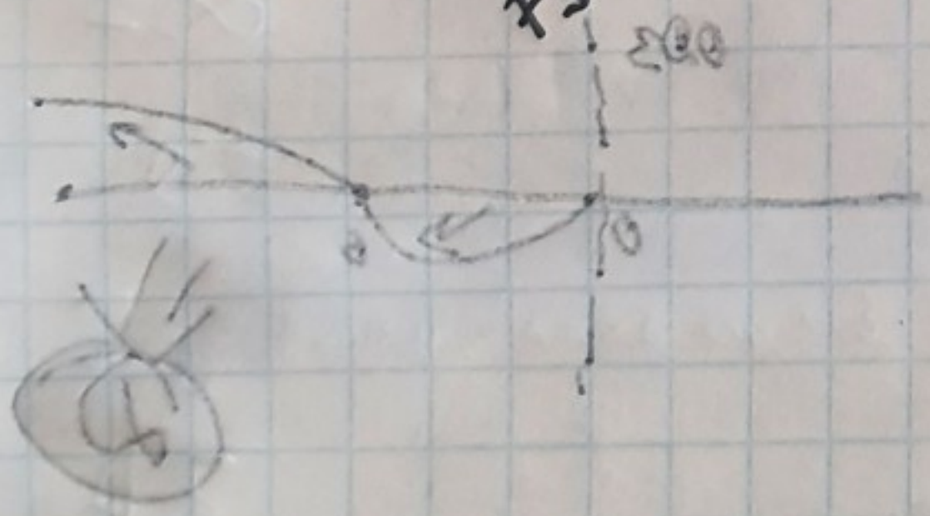
4) $\lim\limits_{x\to 0} \frac{\cos 2x}{x} = \infty \Rightarrow$ zło pogg $\Rightarrow$ (B)

5) $\lim\limits_{n\to\infty} \frac{(\sqrt{n^2+1}+n)^2}{\sqrt[3]{n^6+1}} = \lim\limits_{n\to\infty} \frac{\left(n^2\left(\sqrt{1+\frac{1}{n^2}}+1\right)\right)^2}{n^2\sqrt[3]{1+\frac{1}{n^6}}} =$

$= \lim\limits_{n\to\infty} \frac{\left(\sqrt{1+\frac{1}{n^2}}+1\right)^2}{\sqrt[3]{1+\frac{1}{n^6}}} \Rightarrow \frac{(1+1)^2}{1} = 4$

6) $\lim\limits_{x\to 0} \frac{\sin^2 2x\left(\sqrt[4]{1+x^{\ldots}}-1\right)}{x^3} = \left|\begin{array}{l}\sin 2x \sim 2x \\ \left(1+x^{\ldots}\right)^{\frac{1}{4}} \sim 1+\frac{1}{4}x^{\ldots}\end{array}\right| =$

$= \lim\limits_{x\to 0} \frac{4x^2 \cdot \frac{1}{4}x^{\ldots}}{x^3} = 1 \Rightarrow \ldots = 0$



7) $\lim\limits_{x\to 3} \dfrac{x^2-5x+6}{x^2-9} = \lim\limits_{x\to 3} \dfrac{(x-2)(x-3)}{(x-3)(x+3)} = \lim\limits_{x\to 3} \dfrac{(x-2)}{(x+3)} = \dfrac{1}{6}$

9) $\lim\limits_{n\to\infty} \dfrac{\ln(2+e^{3x})}{\ln(3+e^{2x})} = \left[\dfrac{\infty}{\infty}\right] = \lim\limits_{n\to\infty} \dfrac{\dfrac{3e^{3x}}{2+e^{3x}}}{\dfrac{2\cdot e^{2x}}{3+e^{2x}}} =$

$= \lim\limits_{n\to\infty} \dfrac{3+e^{2x}}{2+e^{3x}} \cdot \dfrac{3}{2} e^{x} = \lim\limits_{n\to\infty} \dfrac{2e^{2x}\cdot 3e^{x} + 3(3+e^{2x})e^{3x}}{6e^{3x}} =$

$= \lim\limits_{n\to\infty} \dfrac{6e^{3x} + 9e^{x} + 3e^{3x}}{6e^{3x}} = \lim\limits_{n\to\infty} \dfrac{9e^{3x} + 9e^{x}}{6e^{3x}} = \lim\limits_{n\to\infty} \dfrac{3}{2} + \dfrac{3}{2e^{2x}} = \dfrac{3}{2}$

8) $\lim\limits_{x\to 2} \left(\dfrac{x}{2}\right)^{\frac{1}{x-2}} = [1^{\infty}] = e^{\lim\limits_{x\to 2} \frac{\ln \frac{x}{2}}{x-2}} = e^{\lim\limits_{x\to 2} \frac{\frac{2}{x}\cdot\frac{1}{2}}{1}} = e^{\lim\limits_{x\to 2} \frac{1}{x}} = \sqrt{e}$

10) $y = e^{3\arcsin x}(x+1)$

$y' = \left(e^{3\arcsin x} \cdot 3 \cdot \dfrac{1}{\sqrt{1-x^2}}\right)(x+1) + 1\left(e^{3\arcsin x}\right) =$

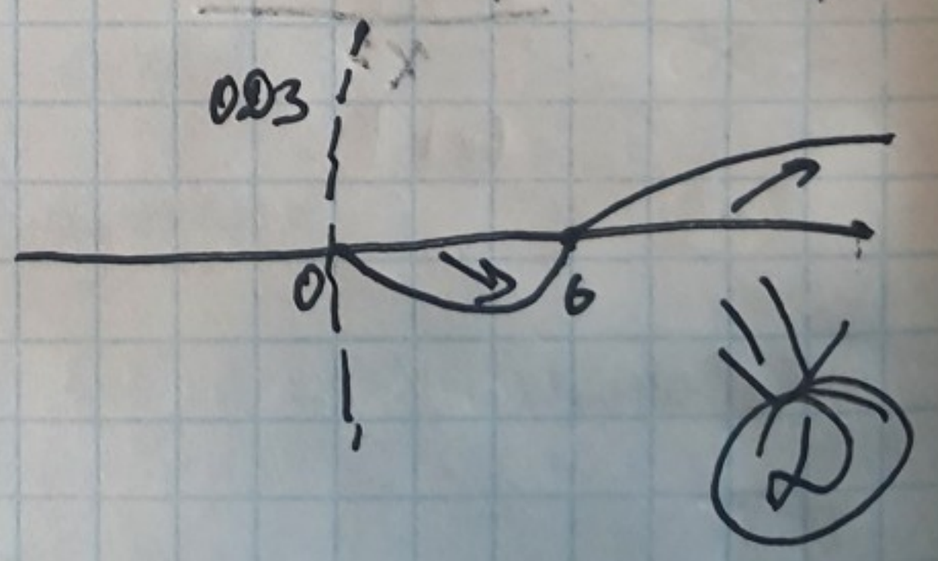
$= \dfrac{3e^{3\arcsin x}(x+1)}{\sqrt{1-x^2}} + e^{3\arcsin x}$

$y'(0) = \dfrac{3e^{0}(1)}{\sqrt{1}} + e^{0} = 3+1 = 4$

12) $y = x^2 - 8x - 24\ln x + 3$ $\quad \ln x \Rightarrow ОДЗ\ x>0$

$y' = 2x - 8 - \dfrac{24}{x} = 0$

$\dfrac{2x^2-8x-24}{x} = 0$



$x=0$
$x=-2$
$x=6$ | ОДЗ $\Longrightarrow$ $x=6$

12) $y = x^3 + 6x\ln x$ ОДЗ: $x > 0$

$y' = 3x^2 + \left(6\ln x + 6\right)$

$y'' = 6x + \frac{6}{x} = \frac{6x^2+6}{x}$ $\forall x: y'' > 0 \Rightarrow$ (D) нет т. перег.

13) $\lim\limits_{x\to 0} \frac{\operatorname{arcsin}\sqrt{x}}{\sqrt{1-x}-1} = \left[\frac{0}{0}\right] = \lim\limits_{x\to 0} \frac{\frac{1}{\sqrt{1-x}}\cdot\frac{1}{2\sqrt{x}}}{\frac{-1}{2\sqrt{1-x}}} = \lim\limits_{x\to 0} \frac{-1}{\sqrt{x}} = -\infty$

14) $y = (\ln x)^{\frac{1}{x}}$

Далее $y' = \frac{1}{x}\cdot(\ln x)^{\frac{1}{x}-1}\cdot\frac{1}{x} = \frac{\ln x^{\frac{1}{x}-1}}{x^2}$

$y'(e) = \frac{(\ln e)^{\frac{1}{e}-1}}{e^2} = \frac{1^{\frac{1}{e}-1}}{e^2} = \frac{1}{e^2} = e^{-2}$

15) $y = 5x + \frac{3}{x^2-9}$ $x \neq \pm 3$

Вертикальные:

$\lim\limits_{x\to -3-0} 5x + \frac{3}{x^2-9} = \infty \Rightarrow$ Разрыв 2го рода $\Rightarrow x = -3$ — верт ас.

$\lim\limits_{x\to -3+0} 5x + \frac{3}{x^2-9} = \infty$

$\lim\limits_{x\to 3-0} 5x + \frac{3}{x^2-9} = \infty \Rightarrow$ Разрыв 2го рода $\Rightarrow x = 3$ — верт ас.

$\lim\limits_{x\to 3+0} 5x + \frac{3}{x^2-9} = \infty$

Горизонтальні:

$\lim\limits_{x\to\infty} 5x + \frac{3}{x^2-9} = [\infty + 0] \neq \infty$

$\lim\limits_{x\to-\infty} 5x + \frac{3}{x^2-9} = [-\infty + 0] = -\infty$

} Горизонтальних асимптот немає

Похилі:

$$\lim_{x\to\pm\infty} \frac{5x + \frac{3}{x^2-9}}{x} = \lim_{x\to\pm\infty} 5 + \frac{3}{x(x^2-9)} = 5 = k$$

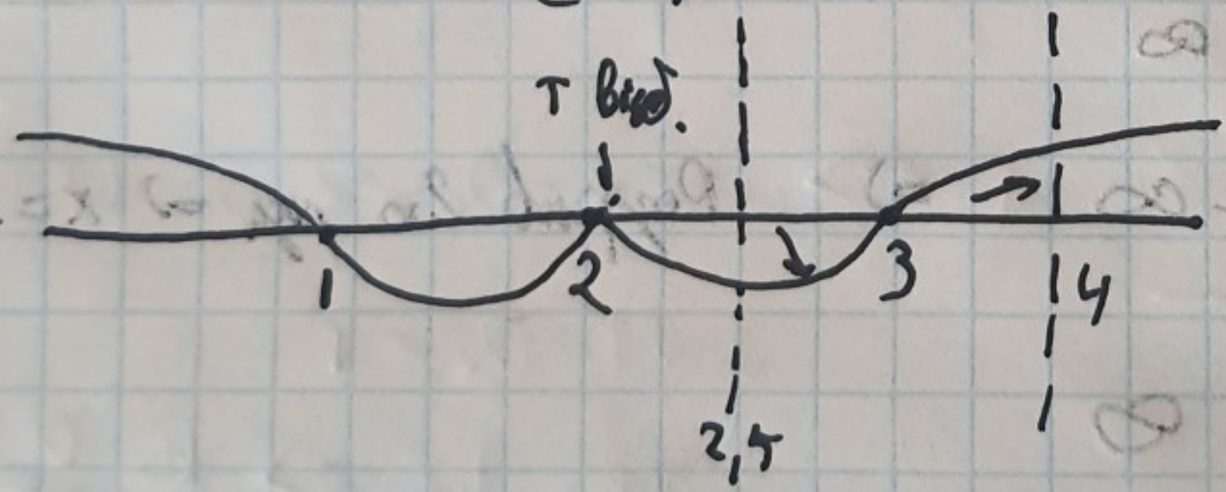
$$\lim_{x\to\pm\infty} 5x + \frac{3}{x^2-9} - 5x = \lim_{x\to\infty} \frac{3}{x^2-9} = 0 \Rightarrow$$

$\Rightarrow$ Похила асимптота $y = 5x$

16) $y = \frac{x^2-4x+5}{x-2}$

$$y' = \frac{(2x-4)(x-2) - x^2+4x-5}{(x-2)^2} = \frac{2x^2-8x+8-x^2+4x-5}{(x-2)^2} = \frac{x^2-4x+3}{(x-2)^2}$$

$$y' = 0 = \frac{(x-1)(x-3)}{(x-2)^2} \Rightarrow \begin{array}{l} x=1 \\ x=3 \\ x=2 \end{array}$$



$f(2,5) = \frac{6,25-5}{\frac{1}{2}} = 2,5$

$f(3) = \frac{9-12+5}{1} = 2 = m$

$f(4) = \frac{16-16+5}{2} = 2,5 = M$

$m+M = 2,5+2 = 4,5$